FINAL PRESENTATION

DESIGN OF EXPERIMENTS

Effect of Keyboard Type & Environment Noise on Typing Speed

A 2×2 factorial study of keyboard type and background noise on net typing speed (WPM) — final results from 24 trials across 6 participants

Keyboard Type

Environment Noise

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Central Question

Does keyboard type and noise change how fast people type?

We tested whether keyboard type (laptop built-in vs. external) and background noise (quiet vs. noisy) affect a person's net typing speed.

Keyboard Type

Does an external keyboard type faster than a laptop's built-in one?

Environment Noise

Does background noise slow typing speed down?

Why it matters: most people type on whatever's in front of them — knowing what actually slows them down has real workplace relevance.

How we measured typing speed

Response Variable

WPM

net words per minute, typos corrected live

Y = net WPM, measured once per participant per treatment combination.

How it was measured

- Each trial ran for a fixed 3-minute timed typing test.
- Typos were fixed in real time while typing, so WPM reflects final, error-free text — not raw keystrokes.
- Noisy trials used heavy metal music played at the same volume every time, to maintain a consistent noise condition across all trials.

Two factors, two levels each (2^2 design)

Factor	Level 1	Level 2
A: Keyboard Type	Laptop built-in keyboard	External keyboard
B: Environment Noise	Quiet room	Noisy (fixed audio clip, set volume)

Both factors were set directly by us — the laptop is placed into each condition on purpose, not observed naturally. That means only the treatment, not randomness in setup, drives the result.

Treatment Combinations

$2^2 = 4$ scenarios

Combo	Keyboard (A)	Environment (B)
1	Laptop Keyboard	Quiet
2	Laptop Keyboard	Noisy
3	External Keyboard	Quiet
4	External Keyboard	Noisy

6

participants

× 4 combinations = 24 total trials

Each of the 6 participants completed all 4 rows — participants act as replicate blocks.

Controlling for learning effects, replicating across participants

Randomization

Each participant completed the 4 combinations in an independently randomized order, so no combination was unfairly helped by practice.

Replication

6 participants each completed all 4 combinations — 24 total trials and 6 replicate measurements per treatment. Participants act as blocks, so individual typing-skill differences don't bias the comparison.

Procedure

What happened in each trial

1

Assign order

Randomly generate that participant's order for the 4 treatment combinations.

2

Set condition

Set the keyboard and the quiet/noisy environment for that trial.

3

Run test

Type for a fixed 3 minutes, including fixing typos

4

Record & reset

Record net WPM, then reset before the next trial.

Hypotheses

What we are testing

Three hypotheses, tested at α = 0.05

Keyboard

H0: Mean WPM is equal for Internal and External keyboards.

Ha: Mean WPM differs between keyboard types.

Environment

H0: Mean WPM is equal in Quiet and Noisy conditions.

Ha: Mean WPM differs between Quiet and Noisy.

Interaction

H0: The keyboard difference is the same in both environments.

Ha: The keyboard difference depends on the environment.

The ANOVA model

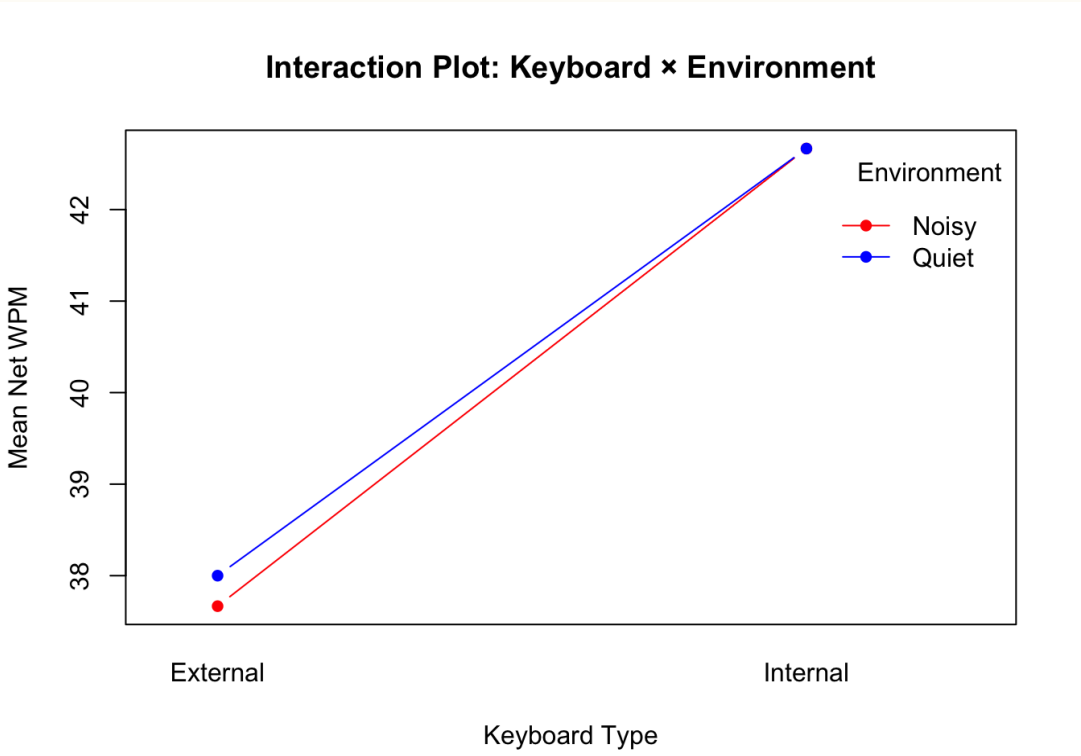
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Net WPM ~ Participant + Keyboard × Environment
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Keyboard & Environment (main effects): test the overall difference between keyboard types, and between quiet and noisy.
Keyboard × Environment (interaction): tests whether the keyboard effect changes depending on the noise level.
Participant (block): accounts for baseline typing-skill differences, since each participant completed all four conditions.

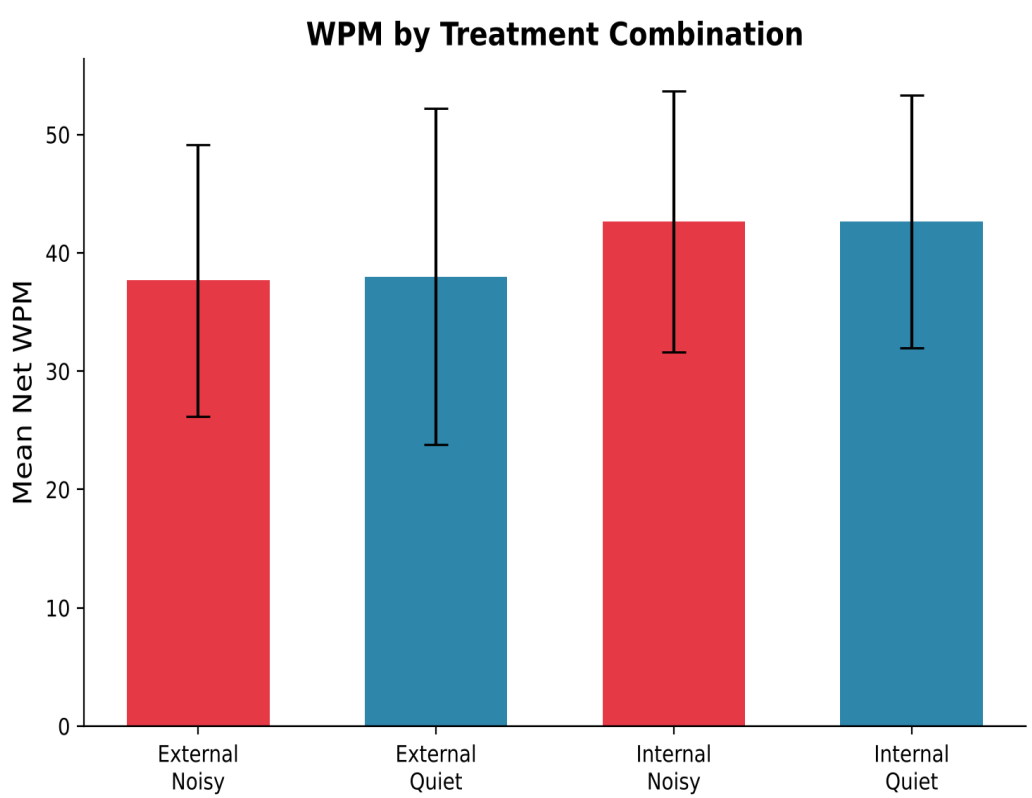
ANOVA Results

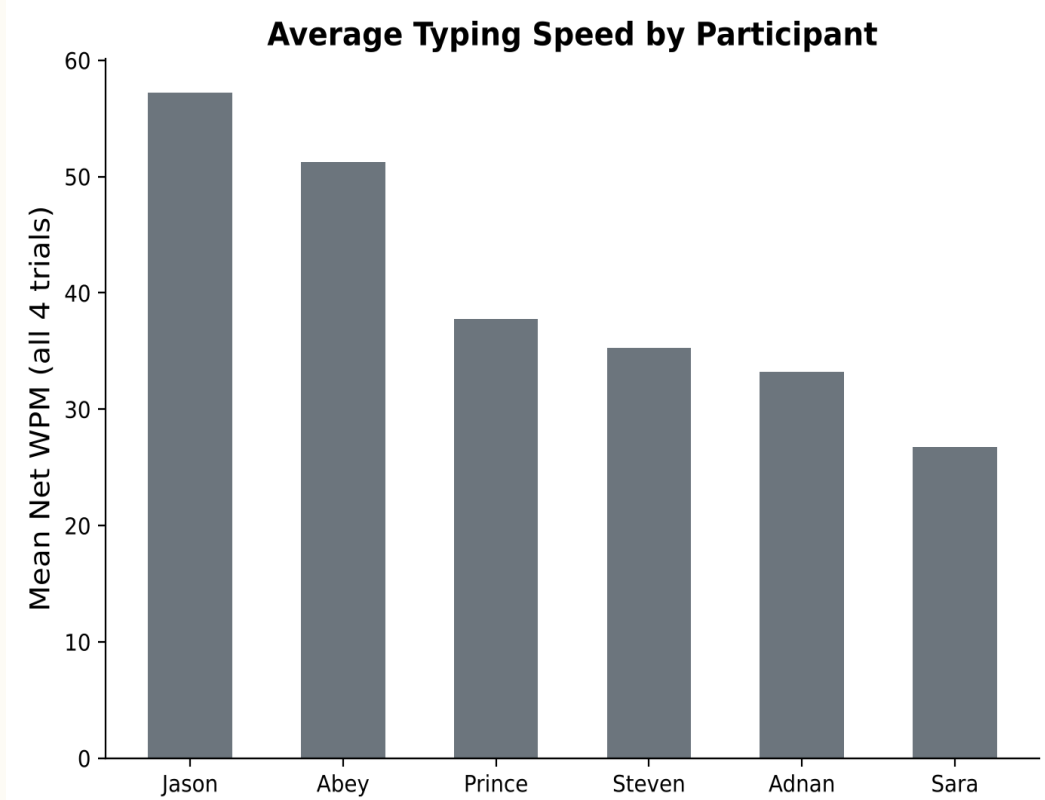
Source	F	p-value	Conclusion
Keyboard	**13.31**	**0.0024**	**Significant**
Environment	0.02	0.902	Not significant
Keyboard × Environment	0.02	0.902	Not significant
Participant (block)	51.08	<0.001	Significant

A p-value below 0.05 means the effect is unlikely to be due to chance alone — at α = 0.05, we reject H0 for Keyboard.



24 trials, 6 participants





Individual participants vary far more than keyboard or environment do — this is why we blocked on participant.

What the data show

At α = 0.05, keyboard type significantly affects typing speed (p = 0.0024). Noise (p = 0.902) and the interaction (p = 0.902) do not.

- Internal (laptop) keyboard averaged faster typing than the external keyboard (42.7 vs. 37.8 WPM), a difference of about 4.8 WPM — a real, statistically significant effect.
- Background noise made almost no difference (40.2 vs. 40.3 WPM) — people typed about the same whether the room was quiet or noisy.
- Participant was by far the strongest source of variation ($p < 0.001$) — individual typing skill matters more than either factor tested.
- Practical takeaway: for this task, which keyboard you use matters more than how noisy the room is.

Retrospective

What we'd change, and biggest challenges

Biggest Challenge

- We corrected a data-entry mistake before analysis — that's how we accounted for error: catching and fixing it, not smoothing over it.
- Coordinating 24 timed trials across 6 people's schedules.

If We Ran It Again

- Recruit even more participants to get a tighter estimate of the keyboard effect size.
- Add 2-3 replicate trials per person per combination to separate trial-to-trial noise from true treatment effects.